

4(a)	<i>any one from:</i> <i>idea that</i> to provide cells that can, differentiate / divide ; for (continued) growth of the, shoot tip / (named) tissues ; ! 'to produce more meristematic cells'	1
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Question	Answer	Marks								
4(b)	<i>one mark per row</i> <table><tr><th>stages</th><th>description</th></tr><tr><td>F and G</td><td>cell, elongates / enlarges / grows and, a vacuole forms / makes a tonoplast ;</td></tr><tr><td>G and H</td><td>cell divides (longitudinally) and one of the cells has a vacuole ;</td></tr><tr><td>H and J</td><td>developing sieve tube cell, elongates / enlarges / grows and loses its nucleus or one (developing companion) cell divides (transversely) to form (two) companion cells ; A companion cell divides</td></tr></table>	stages	description	F and G	cell, elongates / enlarges / grows and , a vacuole forms / makes a tonoplast ;	G and H	cell divides (longitudinally) and one of the cells has a vacuole ;	H and J	developing sieve tube cell, elongates / enlarges / grows and loses its nucleus or one (developing companion) cell divides (transversely) to form (two) companion cells ; A companion cell divides	3
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4(c)	<i>any four from:</i> 1 elongated cells to form, long tubes / AW ; 2 little, cytoplasm / cell contents / fewer organelles, to reduce resistance to flow ; A peripheral cytoplasm / no nucleus, to allow transport of maximum volume of, (named) assimilates / sap / nutrients A ‘more space for’ R no organelles R ‘no cell contents’ 3 sieve (plates have) pores, so little barrier to flow from cell to cell / easy (for phloem sap) to pass from cell to cell / allows mass flow ; 4 sieve plates, support / stop collapse of / stop bulging of, sieve tube elements ; A become plugged with, P-protein / callose, to prevent losses / after damage A maintain hydrostatic pressure (in sieve tubes) 5 <u>plasmodesmata</u> between sieve tube and companion cell for ease of, loading / unloading / AW ; 6 AVP ; e.g. membrane around sieve tube for osmosis to occur / prevent loss of (named) assimilates	4								

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4(d)	<i>allow assimilates / AW for sucrose</i> 1 movement of sucrose from, mesophyll / parenchyma / source, cells ; A movement of sucrose to (named) sink cells 2 move sucrose, into / out of, sieve tubes (through plasmodesmata) ; 3 pump, protons / hydrogen ions / H⁺, out of cell / into cell wall/into apoplast ; A <i>ref. to</i> secondary active transport 4 provides, ATP / proteins, for sieve tubes ; A maintain metabolism of sieve tubes	2

Question	Answer	Marks
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